Capgemini

AWS Global
Agentic AI Hackathon

Team:
Digital Hackers

In collaboration with aws onova

C4: Problem Statement

Restaurants face challenges in maximizing capacity while delivering personalized guest experiences due to fragmented or manual reservation and table management systems. This use case proposes a **multi-agent AI system** to streamline operations and enhance service quality.

HOW MIGHT WE USE AI AGENTS TO HELP WITH:

- ✓ Reservation Management
- ✓ Table Optimization
- ✓ Guest Personalization
- ✓ POS Integration with CRM & Kitchen Operations to enable seamless front-of-house orchestration

WITH THESE PARAMETERS IN MIND…

- Real-time data ingestion (seating, capacity, reservations)
- No-show prediction and waitlist auto-fill
- Smart table combinations for groups
- Personalized communication and suggestions
- Adaptive turnover strategies based on live traffic
- Integration with POS, CRM, and kitchen systems
- Performance tracking (wait times, utilization, satisfaction)

Target Customers

An Agentic AI solution like this targets the following customers/ users:

Restaurant Owners & Operators

- Independent restaurants – looking to improve efficiency & guest satisfaction
- Restaurant Groups or Chains – aiming to standardize & scale operations

Hospitality Management Companies

- Companies that manage multiple restaurant brands or venues and need a centralized, intelligent system to optimize performance

Front of House Staff

- Hosts & managers who benefit from automated waitlist handling, table assignments and guest communications
- Servers who receive guest insights (e.g. preferences, allergies) to personalize service

Marketing & CRM Teams

- Use guest data & AI-driven insights to run targeted campaigns, loyalty programs and personalized promotions

Event Planners & Coordinators

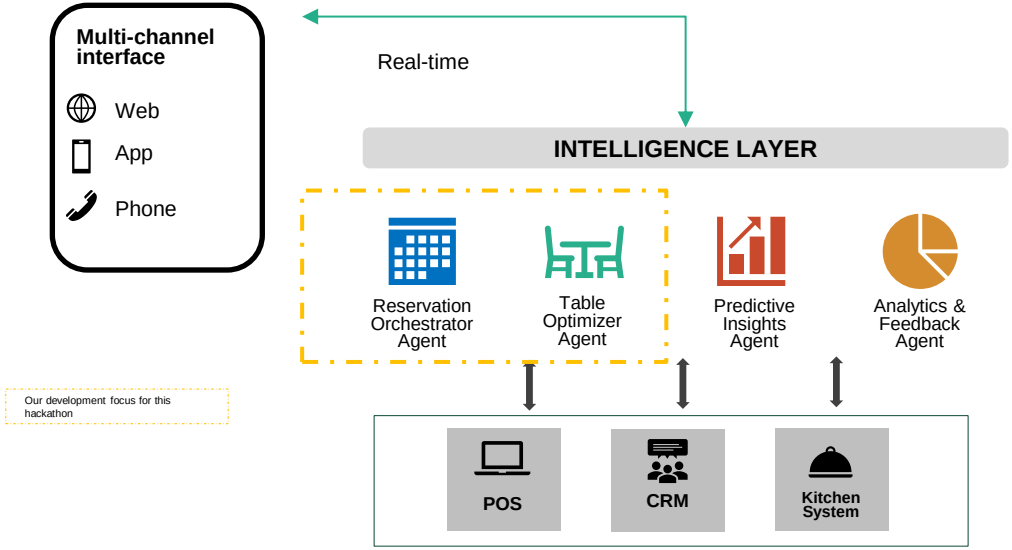
- For venues that host private events or large parties, this solution helps with space planning & guest experience management

Technology Vendors & Integrators

- POS Providers, CRM platforms & kitchen display system vendors who can integrate with or resell the AI solution

Solution Overview

Our Agentic AI solution is summarized as follows:



Hardware, Software, APIs in the solution:

XXX

Solution Deep Dive:

Reservation Orchestrator Agent – the central coordinator for restaurant operations

- Real-time data ingestion - pulls data from multi-channel interfaces (reservations); live seating layout; reservation database
- Waitlist & auto-fill – monitors cancellations & no-shows (from predictive insights agent); prioritizes based on loyalty status and party size, preferences
- Coordinates with other agents – e.g. table optimization

Table Optimization Agent

- Table Matching & combinations – matches parties to optimal tables based on size, guest preferences (quiet, window), turn-over timing
- Combines adjacent tables for large parties
- Turn-over prediction – based on party size, meal order (tasting menu vs. quick lunch), historical data
- Dynamic reallocation – party is running late, no show confirmed, VIP arrives

Predictive Insights Agent

- No-show & cancellation predictions
- Demand forecasting

Analytics & Feedback Agent

- Real-time metrics tracking (table turn-over rate, average wait time, guest satisfaction NPS, capacity utilization)
- Feedback loop to reservation (adjust booking strategies), table optimization (improve seating logic) & predictive insights (refine no-show models) agents

Technical Demo

Key Improvement Metrics

Based on benchmarks/ industry sources*, multi-agent AI restaurant solutions like this could drive a range of improvements:

	Improvement
Operational Efficiency	• 10% - 25% increase table turnover rate • 20%-40% reduced average wait time • 30%-50% reduction in no show rates • 50%-70% reduction in manual interventions
Revenue & Upsell	• 10% - 20% increase in revenue per available seat hour • 15%-30% increase in upsell conversion rate • 20%-40% increase in large party accommodation
Guest Experience Metrics	• 10% - 25% increase in guest promoter scores • 15%-30% increase in repeat visit rate • 20%-40% personalized experience scores

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* Optivuze; Livelytics

Timeline & Execution Plan

What is your **execution plan**?

How will you *reach* your users/customers (web, direct, phone, etc.)?

What is the **timeline**?

THANK YOU